

Cryptography & SSL/TLS Testing Lab Report

1. Introduction

SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are essential protocols used to secure communication between a user and a web server. They ensure that the data transmitted over the internet is encrypted and protected from attackers. However, many websites still use weak encryption protocols, expired certificates, or insecure configurations, which can lead to serious security risks such as Man-in-the-Middle (MITM) attacks. This project focuses on analyzing SSL/TLS configurations of websites to identify security weaknesses such as expired certificates, weak ciphers, and outdated TLS versions.

2. Abstract

This project presents a Cryptography & SSL/TLS Testing Lab designed to analyze the security configuration of websites. The system scans a given domain and evaluates its SSL/TLS implementation to detect potential vulnerabilities. It identifies issues such as expired SSL certificates, weak encryption protocols, and possible Man-in-the-Middle (MITM) risks. The tool automatically retrieves SSL certificate information and verifies its validity period. It also checks the TLS version used by the server to determine encryption strength. This project helps students and security testers understand SSL/TLS security and perform basic vulnerability assessments.

3. Tools Used

Python, Flask, SSL Library, Socket Programming, Web Browser, OpenSSL, Nmap.

4. Steps Involved in Building the Project

1. Install Python and required libraries such as Flask, SSL, and socket. 2. Create a Flask web application. 3. Design a web interface where users can enter a domain name. 4. Establish a secure connection with the server using SSL and socket libraries. 5. Retrieve SSL certificate details from the server. 6. Check certificate validity and expiration date. 7. Identify TLS version and encryption strength. 8. Display the results on the web interface. 9. Analyze whether the website is secure or vulnerable.

5. Conclusion

The Cryptography & SSL/TLS Testing Lab helps in analyzing the security configuration of websites and identifying vulnerabilities related to SSL/TLS. The project successfully detects expired certificates, weak TLS versions, and possible MITM risks. This project improves understanding of web security and encryption protocols and helps in performing basic security testing. It can be further improved by adding automated vulnerability reports and support for scanning multiple domains.